Study Coach AI

Milestone 1: AI-Ideated Concepts

HAI Principle: User Control and Agency

Project Track: Study Coach and Exam Prep

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Purpose

To explore and compare two AI-generated concepts for the core interaction flow of Study Coach AI and select a suitable direction for the project.

1. Concept Draft 1 — ChatGPT

1.1 Tool Information

- **AI Tool:** ChatGPT
- **Purpose:** To generate a concept for the core interaction flow of Study Coach AI.
- **HAI Principle:** User Control and Agency
- **Date:** October 10, 2026

1.2 Prompt Used

Prompt 1 :

We are a team of four university students working on a Human-AI Interaction (HAI) term project called "Study Coach AI".

The project aims to help university students study course materials, practice exam questions, understand their mistakes, and prepare for exams.

Our selected HAI principle is **User Control and Agency**. Students should be able to choose the type and level of help they receive from the AI.

Propose one detailed concept for the core interaction flow of this chatbot.

Please include:

1. The steps from opening the chatbot to completing a study task.
2. How students select between a hint, a step-by-step explanation, and a full answer.
3. How the AI helps students understand mistakes.
4. How students can change their preferences or correct the AI.
5. What the chatbot should do when it does not know the answer or lacks reliable course information.
6. How this concept supports User Control and Agency.

Keep the concept realistic for a student team to implement as a web-based prototype. Present the interaction as numbered steps and include one short example conversation.

1.3 Original AI Output

Study Coach AI — Core Interaction Flow Concept

Concept overview

Study Coach AI is a chatbot that helps university students study course materials, practice exam questions, and learn from mistakes. Instead of automatically giving students answers, it lets them decide how much assistance they want at every stage.

The main idea is: the student controls the learning process, and the AI supports their decisions rather than making them on their behalf.

1. Core interaction flow

Step 1 — Open the chatbot

The student opens the Study Coach AI website and sees a simple dashboard with three options: Study a topic, Practice exam questions, and Review my mistakes.

User control: The student chooses the task instead of being forced into a predefined learning path.

Step 2 — Select the course material

The student selects a course and a topic, then uploads a PDF or selects material already available in the prototype. If no material is uploaded, the chatbot explains that its answers may not match the student's course exactly.

User control: The student decides what to study and which material the AI should use.

Step 3 — Choose the study mode

The student chooses between learning a concept, solving a practice question, or reviewing a previous mistake. They can also select their preferred assistance level.

User control: The student determines how the interaction will proceed.

Step 4 — Receive assistance at the chosen level

The chatbot responds with a hint, a step-by-step explanation, or a full answer. The student can request a different level at any point without restarting the task.

User control: Help is adjustable, not imposed by the AI.

Step 5 — Submit an answer and review mistakes

For practice questions, the student submits their answer. The chatbot compares it with a reference answer or a clearly defined rubric, identifies potential errors, and explains the relevant concept.

User control: The student can ask for clarification, challenge the feedback, or try again.

Step 6 — Complete the task

The chatbot asks whether the student wants to retry, practice a similar question, review the explanation, or finish the session. A short summary shows what was completed and which concepts may need more practice.

User control: The student chooses the next step or ends the session at any time.

2. How students choose the level of AI assistance

Whenever the student asks a question or gets stuck, the chatbot displays three clear options. It should not reveal the full solution automatically when the student requests a hint.

A. Hint

Gives a small clue, a relevant formula, or a guiding question without revealing the complete solution.

Example: “Which condition must be true before you apply this formula?”

B. Step-by-step explanation

Breaks the solution into manageable steps, explaining why each step is necessary.

Example: “First, identify the known values. Next, select the formula. Finally, substitute the values and simplify.”

C. Full answer

Provides the complete solution, including the reasoning and final result.

Example: A complete worked solution that the student can compare with their own attempt.

The student can switch between these options using buttons below the response. For example, selecting Hint and then Show step-by-step should expand the assistance without deleting the previous conversation.

The chosen assistance level can remain active for the current session, but it should always be easy to change.

3. How the AI helps students understand mistakes

When a student submits an incorrect answer, the chatbot should do more than mark it wrong. It should provide four things:

1. Identify the error: Point out the specific step, concept, or assumption that appears incorrect.
2. Explain why: Describe the underlying misconception in accessible language.
3. Offer a correction: Let the student choose a hint, a step-by-step explanation, or a full corrected solution.
4. Encourage another attempt: Offer the opportunity to revise the answer or try a similar question.

For example, if a student uses the wrong formula, the chatbot could say:

> “Your substitution is correct, but the formula you selected applies to a different case. Would you like a hint to identify the appropriate formula, or would you prefer a step-by-step explanation?”

The chatbot should use a reference answer or rubric when available. If it cannot confidently determine whether the student's answer is wrong, it should explain the uncertainty rather than falsely correcting the student.

4. Changing preferences and correcting the AI

The interface should provide a small Preferences menu and response-level controls.

Feature

Student action

Expected behavior

Assistance level

Change Hint to Full Answer

The next response follows the new level.

Explanation style

Choose concise or detailed

The AI adjusts the amount of explanation.

Learning pace

Choose one step at a time or all steps

The AI presents information accordingly.

Correct the AI

Select “This seems incorrect”

The AI asks what seems wrong and rechecks against the course material.

Source correction

Point to a relevant PDF page or section

The AI uses that reference to reconsider its answer.

Undo or retry

Request another explanation or retry a question

The student can recover from an unhelpful interaction without restarting.

The student should also be able to dismiss a suggestion, skip a question, or end the session. These controls should be visible and straightforward rather than hidden in complex menus.

5. What happens when the AI does not know?

Study Coach AI should have an explicit uncertainty-handling process.

1. Check available evidence. Determine whether the uploaded course material contains relevant information or whether a reliable reference answer is available.
2. Explain the limitation. If the material is missing, ambiguous, or insufficient, tell the student what is uncertain.
3. Offer choices. Ask the student to upload the relevant lecture, share a page number, provide additional context, or request a general explanation clearly labeled as such.
4. Avoid fabricated answers. Do not invent quotations, page references, course rules, or supposedly verified solutions.

Example message:

> “I couldn't find enough information in your uploaded lecture to verify this answer. You can upload the relevant page, ask me for a general explanation that may not match your course exactly, or skip this question.”

This approach preserves student agency by allowing the student to decide how to proceed instead of silently accepting an unreliable answer.

6. Short example conversation

Student

I don't understand how to solve this question about sets. Can you help me?

Study Coach AI

Of course! How would you like me to help?

Give me a hint

Explain it step by step

Show me the full answer

Continue with my choice

Student

Give me a hint first.

Start by identifying which elements belong to both sets. Remember that the intersection contains only the elements shared by the two sets.

Would you like to try the question now, or see a step-by-step explanation?

Student

I tried it, but I think I made a mistake.

No problem. Share your answer and I'll help you identify the step that may be incorrect. You can choose whether to receive another hint or see the complete explanation.

7. Why this concept supports User Control and Agency

The concept applies the selected HAI principle in six practical ways:

- Choice: Students choose the task, course material, and assistance level.
- Adjustability: They can change how the AI helps at any point.
- Transparency: The chatbot explains uncertainty and indicates when course information is unavailable.
- Correctability: Students can challenge AI responses and provide better source material.
- Reversibility: Students can retry, request another explanation, or change direction without restarting.
- Ownership: Students decide whether to attempt a question independently, receive guidance, or view the full answer.

The key design decision is that the AI should offer assistance without taking control away from the student.

8. Realistic implementation for a team of four

The prototype can be built as a simple web application without training a custom AI model.

Team member	Main responsibility
Student 1	UI/UX design, chatbot interface, assistance-level controls
Student 2	Frontend development and interaction flow
Student 3	AI integration, prompts, and response handling
Student 4	

Course-material handling, testing, and evaluation of user control

Suggested prototype features: a chat interface, PDF upload or sample course material, three assistance buttons, a practice question with feedback, a preferences menu, and an uncertainty message.

To keep the project manageable, start with one course and a small set of verified questions. Use an existing language-model API and basic document retrieval if time permits. Store preferences in the session and use predefined test cases to evaluate whether the AI respects the student's choices.

Success criterion: A student should be able to complete a study task while remaining in control of the amount of help received, the next action, and whether to accept or challenge the AI's feedback.

2. Concept Draft 2 — Google Gemini

2.1 Tool Information

- **AI Tool:** Google Gemini
- **Purpose:** To generate an alternative concept for the core interaction flow of Study Coach AI.
- **HAI Principle:** User Control and Agency
- **Date:** October 10, 2026

2.2 Prompt Used

We are a team of four university students working on a Human-AI Interaction (HAI) term project called "Study Coach AI".

The project aims to help university students study course materials, practice exam questions, understand their mistakes, and prepare for exams.

Our selected HAI principle is **User Control and Agency**. Students should be able to choose the type and level of help they receive from the AI.

Propose one detailed concept for the core interaction flow of this chatbot.

Please include:

1. The steps from opening the chatbot to completing a study task.
2. How students select between a hint, a step-by-step explanation, and a full answer.
3. How the AI helps students understand mistakes.
4. How students can change their preferences or correct the AI.
5. What the chatbot should do when it does not know the answer or lacks reliable course information.
6. How this concept supports User Control and Agency.

Keep the concept realistic for a student team to implement as a web-based prototype. Present the interaction as numbered steps and include one short example conversation.

2.3 Original AI Output

1. Core Interaction Flow: Step-by-Step

[Upload & Goal] ➔ [Practice Question] ➔ [Select Assistance Level] ➔ [Student Answer / Review] ➔ [Correction & Preference Control]

1. **Upload & Goal Setting:** The student opens the web interface and uploads study materials (e.g., lecture slides, syllabus, or PDF notes). They select their goal (e.g., "Prepare for Midterm", "Quick Concept Review", or "Practice Exam Questions").
2. **Question Generation & Mode Selection:** The AI generates or retrieves a practice question aligned with the course content.
3. **Scaffolding Controls (Assistance Selection):** Before or during their answer attempt, the student is presented with explicit buttons to control the level of AI aid:
    - Get a Hint (Minimal aid)
    - Step-by-Step Breakdown (Guided aid)
    - Reveal Full Answer & Explanation (Direct aid)
4. **Student Attempt & Error Diagnostic:** The student submits their response or requests feedback. If the answer is incorrect or partial, the AI pinpoints the mistake without being overly prescriptive, giving the user agency over how much deeper to explore.
5. **Feedback & Preference Adjustment:** At any point, the student can adjust the AI's coaching style, correct mistaken assumptions, or flag unhelpful responses.

2. Selecting Assistance Levels

To ensure high user agency, help controls are **always visible and accessible**:

- **Action Toolbar:** Below each question, a three-tier "Help Toolbar" allows instant selection:
    - **Level 1 (Hint):** Offers a conceptual clue or points to the relevant section/formula without solving any part of the problem.
    - **Level 2 (Step-by-Step):** Breaks the problem into sub-questions or sequential steps, guiding the student to solve one step at a time interactively.
    - **Level 3 (Full Answer):** Displays the full solution, key key concepts, and standard working.
- **Dynamic Mode Switcher:** Students can change their default mode (e.g., *"Socratic Mode"* where hints are offered first, vs. *"Direct Answer Mode"*) via a quick toggle in the upper corner of the chat window.

3. How AI Helps Students Understand Mistakes

When an answer is incorrect, the AI uses **Targeted Mistake Analysis**:

1. **Categorized Diagnosis:** The AI classifies the error into distinct categories (e.g., *Conceptual Misunderstanding*, *Calculation Error*, or *Incomplete Step*).
2. **Highlight & Comparison:** The response highlights the exact line/phrase where the logic diverged, showing:
    - **What you wrote:** *Logic A*
    - **Where the gap is:** *Why Logic A doesn't apply under this condition*
3. **Guided Self-Correction:** Instead of just giving the correct answer, the AI asks a targeted follow-up question to help the student spot their own mistake, accompanied by an option to reveal the fix if stuck.

4. Student Preferences & Correcting the AI

- **Live Persona & Style Toggles:** A sliding preference panel lets students set parameters anytime:
    - **Pacing:** *Fast-paced / Detailed / Conversational*
    - **Detail Level:** *Bullet points / Deep dive / Concise*
- **"Correct the AI" Button:** If the AI interprets a course rule incorrectly or hallucinates:
    - Students can click Override AI / Provide Context directly on any message.
    - They can type: *"In our lecture, the professor defined X differently"* or paste a text snippet from their notes.
    - The AI immediately acknowledges the correction, updates its working context for the session, and re-evaluates its answer.

5. Graceful Handling of Unknowns & Missing Info

When course materials lack context or the AI has low confidence:

1. **Uncertainty Flag:** The AI explicitly states its limitation instead of guessing: *"I don't have enough information from your uploaded course slides to verify the specific formula your professor uses for this topic."*
2. **Actionable Recovery Options:** The AI presents three student-driven choices:
    - **Option A:** *"Paste the specific slide text or formula here so I can learn it."*
    - **Option B:** *"Let me answer using standard textbook concepts (note: may differ from lecture)."*
    - **Option C:** *"Skip this question and move to the next topic."*

6. How This Supports User Control and Agency

- **Pacing and Rigor Control:** Students remain in the driver's seat of their learning journey, choosing whether to struggle productively (Hints/Step-by-step) or review quickly (Full Answer).
- **Transparent Knowledge Bounds:** By explicitly stating uncertainty and allowing manual context overrides, the AI position shifts from an infallible authority to a collaborative tool governed by the student.
- **Low Implementation Overhead:** For a student prototype, this can be built with standard UI components (buttons/toggles) combined with structured prompt engineering (passing user-selected parameters like assistance_level: hint to an LLM API).

Example Conversation

Study Coach AI:

Here is a practice question from **CS101 - Algorithms**:

What is the time complexity of searching for an element in an unbalanced Binary Search Tree (BST) in the worst case, and why?

(Choose your assistance level below: [Hint] [Step-by-Step] [Reveal Answer])

Student:

[Clicks Hint]

Study Coach AI:

Hint: Think about what an *unbalanced* BST looks like when elements are inserted in sorted order. What basic data structure does it resemble?

Student:

Is it $O(\log n)$ because it's a tree structure?

Study Coach AI:

Mistake Analysis: Conceptual Misunderstanding

- **Your answer:** $O(\log n)$
- **The Gap:** $O(\log n)$ applies to a *balanced* BST where the height is kept minimal.

Consider this: If you insert nodes 1 -> 2 -> 3 -> 4 in order into an unbalanced BST, what does the tree shape look like?

(Options: [Try again] | [Step-by-Step] | [Reveal Answer])

Student:

Ah, it becomes a linked list! So it takes $O(n)$ time.

Study Coach AI:

Spot on! In the worst case, an unbalanced BST devolves into a linked list, requiring linear search time $O(n)$.

Ready for the next practice problem or would you like to tweak your review preferences?

Suggested Next Steps

If you'd like, I can help you draft the **system prompts** for the LLM to handle these assistance levels, or outline a simple **frontend/backend architecture** suitable for your web prototype!

2.4 Revised Google Gemini Output: Feedback Workflow

This revised output is added separately. The original Gemini prompt and original output remain unchanged as historical records in Section 2.

Revised core interaction flow

- 1. Upload and set a goal. The student uploads or selects course material and chooses a goal such as exam preparation, concept review, or practice questions.
- 2. Generate or select a question. The chatbot offers a question aligned with the available course material.
- 3. Choose the assistance level. The student selects Hint, Step-by-Step Breakdown, or Full Answer using the visible Help Toolbar.
- 4. Attempt and review. The student answers or requests help. The chatbot responds at the selected level and may identify a likely error when there is enough evidence.
- 5. Give optional response-level feedback. After each AI response, show Helpful, Not helpful, and Skip. If Not helpful is selected, the student may add a short reason.
- 6. Adjust the next response within this session. The chatbot uses the rating and optional reason to shape its next response in the current study session only. It does not save this as a long-term preference, and the student can still change the assistance level directly.
- 7. Adjust preferences or correct the AI. The student can change the coaching style, provide context, or flag an assumption or response that seems wrong.
- 8. Choose the next action. The student can retry, review the explanation, practice a similar question, or finish the session.

Revised Gemini Example Conversation

The feedback controls appear after each AI response. Feedback is optional and applies only to the current study session.

Student: I don’t understand this question about sets. Can you help?

Gemini: Choose how much help you want: Hint, Step-by-Step Breakdown, or Full Answer.

Student: Give me a hint.

Gemini: Start by checking which elements appear in both sets. You can now rate this response: Helpful | Not helpful | Skip.

Student: Not helpful. I need a more specific clue, but not the full answer.

Gemini: Thanks. I’ll make the next hint more specific during this session while keeping your Hint setting. Check one element at a time and ask: does it appear in both sets? You can change to a step-by-step explanation whenever you want.

Student: That helps. I’ll try the first element.

Gemini: Great. After each response, you can mark it Helpful or Not helpful, optionally add a reason, or skip feedback.

Implementation note

For a feasible prototype, the feedback feature can use two rating buttons, an optional short reason, and a session-only adjustment to the next response. The chatbot should not treat a rating as permission to override the student’s explicitly selected assistance level.

Appendix A: AI Prompt Log

Entry A1 — ChatGPT

- **AI Tool:** ChatGPT
- **Date:** October 10, 2026
- **Purpose:** Generate the first concept for the Study Coach AI interaction flow.
- **Exact Prompt:** The full prompt is documented in Section 1.2 — Prompt Used.
- **Output:** The original output is documented in Section 1.3.
- **Team Changes:** The team selected the structured six-step interaction flow, including study-task selection, assistance-level control, mistake review, and task completion.
- **Reason for Changes:** These features provide a clear end-to-end workflow and support User Control and Agency.

Entry A2 — Google Gemini

- **AI Tool:** Google Gemini
- **Date:** October 10, 2026
- **Purpose:** Generate an alternative concept for the same chatbot.
- **Exact Prompt:** The same prompt used for ChatGPT, reproduced in Section 1.2
- **Output:** The original output is documented in Section 2.3.
- **Team Changes:** The team incorporated selected ideas, including visible assistance-level controls and structured mistake feedback, while keeping advanced error classification optional.
- **Reason for Changes:** These choices improve clarity and user control while keeping the prototype feasible for a four-member team.

Entry A3 — Team Synthesis

- **Tools Reviewed:** ChatGPT and Google Gemini.
- **Team Decision:** Use the ChatGPT concept as the primary structure and integrate selected features from Gemini.
- **Changes Made:** Prioritize three assistance levels, transparent uncertainty handling, response correction, and student-controlled next actions.
- **Reason:** To create a manageable prototype that demonstrates User Control and Agency through observable user interactions.

4. Concept Comparison

This section compares the two AI-generated concepts for Study Coach AI. The purpose is to identify the strengths and weaknesses of each concept, evaluate how effectively each supports the HAI principle of User Control and Agency, and select a practical direction for the project prototype.

Both concepts aim to help university students study course materials, practice questions, and understand their mistakes while allowing them to control the type and amount of assistance they receive.

4.1 Quick Side-by-Side Comparison

ChatGPT Concept	Google Gemini Concept
Six-step end-to-end study workflow	Concise interaction sequence
Flexible assistance levels	Visible three-level Help Toolbar
Mistake explanation and retry	Categorized mistake diagnosis
Transparent uncertainty handling	Uncertainty flags and recovery choices
Selected as the primary structure	Selected as a source of additional features

4.2 Comparison Criteria

The two concepts are evaluated using the following criteria:

1. **Clarity and completeness of the interaction flow:** Whether the concept explains the steps students follow from starting a study task to completing it.

2. **User Control and Agency:** Whether students can choose their study mode, assistance level, learning pace, and next action.
3. **Mistake understanding and correction:** Whether the chatbot helps students understand why an answer is incorrect and gives them opportunities to correct it.
4. **Uncertainty and reliability:** Whether the chatbot handles missing course information and uncertain answers transparently.
5. **Personalization and flexibility:** Whether students can adjust the chatbot's explanation style and correct its responses.
6. **Implementation feasibility:** Whether the proposed features can be implemented by a team of four university students within the project's time and technical constraints.

4.3 Strengths and Weaknesses

Concept 1 — ChatGPT

Strengths

- Provides a clearly organized interaction flow with six main steps, from opening the chatbot to completing a study task.
- Gives students three assistance options: Hint, Step-by-Step Explanation, and Full Answer.
- Allows students to change their assistance level without restarting the current task.
- Explains mistake handling as a process that includes identifying the error, explaining its cause, offering a correction, and encouraging another attempt.
- Includes an uncertainty-handling process that encourages the chatbot to use available course materials and acknowledge when the information is insufficient.
- Supports student agency through several forms of control, including changing preferences, challenging AI responses, retrying questions, and ending a session.
- Provides a practical example conversation about set intersection, making the proposed interaction easier to understand.
- Includes an implementation outline and suggested prototype features suitable for a small student team.

Weaknesses

- Some interface controls, such as correcting the AI and selecting source references, need further definition to make their behavior clear and implementable.
- The preferences section contains several options that could increase the complexity of the prototype if all are implemented.
- The concept does not define a detailed method for evaluating whether the chatbot consistently respects the student's selected assistance level.
- Its proposed interface and feature descriptions need to be translated into concrete screens and interactions before development.

Concept 2 — Google Gemini

Strengths

- Presents the interaction as a concise sequence from uploading course materials and setting a goal to practicing, receiving feedback, and adjusting preferences.
- Emphasizes keeping the assistance controls visible through a three-level Help Toolbar.
- Introduces a distinction between Socratic Mode, which guides students through questions, and Direct Answer Mode, which provides more direct explanations.
- Proposes classifying student mistakes into categories such as conceptual misunderstandings, calculation errors, and incomplete steps.
- Suggests comparing the student's response with the expected answer and identifying the specific point where the reasoning went wrong.
- Includes preference controls for pacing and explanation detail, as well as an option to correct or challenge the AI.
- Explicitly addresses missing information through uncertainty indicators and options to provide additional course material, request a general explanation, or skip the question.
- Keeps the central interaction focused on student choice and provides ideas that could improve the prototype's usability.

Weaknesses

- The overall interaction flow is less detailed about how a session ends and what happens after the student completes a task.
- The proposed error-diagnosis features may require more advanced logic than a basic prototype can support reliably.
- The Socratic and Direct Answer modes introduce additional design decisions that the team would need to define and test.
- The concept does not provide as much detail about the complete prototype structure and the division of implementation tasks.
- Some suggested behaviors, such as identifying the exact line where a student's reasoning fails, depend on the question format and may not work consistently for every course or answer type.

Overall Comparison

Criterion	ChatGPT Concept	Google Gemini Concept
Interaction flow	More detailed, with six explicit steps	Concise and focused on the main stages
Assistance-level control	Three levels with switching during a task	Three-level Help Toolbar with visible controls
Mistake analysis	Explains the error and encourages another attempt	Adds error categories and more targeted diagnosis
Uncertainty handling	Checks available evidence and offers alternatives	Makes uncertainty visible and offers alternatives

Personalization	Includes assistance level, explanation style, and pace	Highlights pacing, detail, and interaction modes
Implementation guidance	More explicit about the prototype and team feasibility	Offers useful interaction ideas but needs further scoping

Both concepts support User Control and Agency, but they emphasize different strengths. ChatGPT provides a more complete end-to-end workflow and clearer implementation guidance, while Gemini contributes useful ideas for visible assistance controls, structured mistake diagnosis, and interaction modes.

Neither concept should be adopted without refinement. The selected design should preserve student choice while avoiding features that are too complex to implement or validate within the project constraints.

4.4 Selected Direction

We propose using the ChatGPT concept as the primary structure for the Study Coach AI interaction flow and incorporating selected features from the Google Gemini concept.

This direction is preferred because the ChatGPT concept provides a more complete workflow from the beginning to the end of a study task. The Gemini concept complements it with useful ideas for making assistance controls visible and improving the organization of mistake feedback.

The proposed combined direction includes the following features:

1. **Student-selected study task:** Students can choose to study a topic, practice questions, or review previous mistakes.
2. **Course-material selection:** Students can upload or select course materials. When relevant material is unavailable, the chatbot clearly explains this limitation.
3. **Three assistance levels:** Students can choose Hint, Step-by-Step Explanation, or Full Answer and change their choice at any time.
4. **Optional guided-learning mode:** The chatbot can guide students through questions before revealing the complete answer. This feature will be implemented only if time and resources permit.
5. **Structured mistake feedback:** The chatbot identifies the likely error, explains why it may be incorrect, and offers the student a chance to retry or request more help.
6. **Visible preference controls:** Students can adjust explanation detail and learning pace without restarting the task.
7. **Transparent uncertainty handling:** When the chatbot lacks sufficient course information, it explains the limitation and lets the student provide more material, request a clearly labeled general explanation, or skip the question.
8. **Student-controlled completion:** Students decide whether to retry, practice a similar question, review the explanation, or end the session.

Scope and Implementation Priorities

To keep the prototype realistic for a team of four, the first version will prioritize:

- A basic chatbot interface.
- A small set of verified course materials or practice questions.
- Three visible assistance-level controls.
- Student answer submission and basic mistake feedback.
- A way to challenge an AI response or indicate that it may be incorrect.
- A clear message when the chatbot cannot verify an answer.
- Options to retry or finish a study task.

More advanced features, such as detailed mistake classification, persistent personalization, and a separate Socratic Mode, can be considered optional extensions if time permits.

Justification for the Selected Direction

The combined direction best supports User Control and Agency because it gives students meaningful choices throughout the learning process rather than allowing the AI to determine how much help they receive.

It also balances usability with implementation feasibility. By prioritizing a small set of clearly defined controls and interactions, the team can focus on demonstrating the selected HAI principle instead of attempting to build every suggested feature.

The final design will therefore use the ChatGPT concept's structured workflow, enhanced by selected interaction and feedback ideas from Gemini. The team will refine this direction after reviewing the prototype requirements and receiving feedback.

4.5 Gemini Feedback Update to the Comparison

This supplement updates only the Google Gemini side of the comparison and the related comparison summary. The existing ChatGPT concept and comparison remain unchanged.

Criterion	ChatGPT Concept (unchanged)	Google Gemini Concept (updated)
Interaction flow	More detailed, with six explicit steps.	Concise main stages, now with an explicit response-feedback step.
Assistance-level control	Three levels with switching during a task.	Three-level Help Toolbar with visible controls.
Mistake analysis	Explains the error and encourages another attempt.	Adds error categories and more targeted diagnosis.
Uncertainty handling	Checks available evidence and offers alternatives.	Makes uncertainty visible and offers alternatives.
Personalization	Includes assistance level, explanation style, and pace.	Includes pacing, detail, interaction modes, and session-only use of response feedback.
Implementation guidance	More explicit about the prototype and team feasibility.	Useful interaction ideas; the feedback feature adds a small interface and evaluation step.
Response-level feedback	Not changed in this revision; see the original ChatGPT comparison.	Helpful / Not helpful after a response; optional reason; student may skip; next reply adapts within the current session only.

Updated Gemini strengths

- The response-level rating gives students a direct way to signal whether an answer helped.
- An optional reason gives the chatbot context for the next response.
- Students can skip feedback, and the signal affects only the current study session.
- The visible assistance choices and uncertainty options in the original Gemini concept remain useful.

Updated Gemini weaknesses and limits

- The feedback control adds a small interface branch and a behavior the team must evaluate.
- A short reason may be ambiguous, so it should guide the next reply without overriding the student’s selected assistance level.
- The existing concerns about the effort and reliability of detailed error diagnosis still apply.

Updated comparison summary

ChatGPT still offers the more detailed end-to-end workflow and clearer implementation guidance. Gemini still contributes visible assistance controls and structured mistake feedback; the revised Gemini draft now also includes response-level feedback with optional, session-only adaptation. The existing selection of ChatGPT as the primary structure is unchanged. The Gemini feedback step can be considered as an additional feature if the team confirms it fits the prototype scope.

5. Final Proposal for Approval

5.1 Project Title

Study Coach AI: A Student-Controlled AI Study Assistant

5.2 Problem Statement

University students often face difficulties understanding course materials, practicing exam questions, and identifying the reasons behind their mistakes. Traditional study methods may not always provide immediate, personalized feedback, while AI chatbots may reveal complete answers too quickly or provide explanations that do not match the student's course materials.

These challenges can reduce students' opportunities to practice independently and make their own learning decisions.

Study Coach AI is a web-based chatbot concept designed to help university students study course materials, practice questions, understand mistakes, and prepare for exams. Its main objective is to allow students to control how they interact with the AI by choosing their preferred assistance level, adjusting explanations, challenging potentially incorrect responses, and deciding what to do next.

The project focuses not only on providing useful educational assistance but also on ensuring that students remain active participants in their own learning process.

5.3 Target Users

The primary target users are undergraduate university students studying introductory Computer Science courses, particularly students who need help understanding theoretical concepts and practicing exam questions.

Study Coach AI focuses on students who want to control how much assistance they receive while studying. Instead of automatically providing complete solutions, the chatbot allows them to choose between hints, step-by-step explanations, and full answers.

The project differs from a general-purpose educational chatbot by making student control over assistance level, mistake correction, and learning decisions the central part of the interaction design.

The initial prototype will focus on a limited set of questions from one selected Computer Science course.

5.4 Proposed Core Interaction Flow

The proposed interaction flow combines the structured workflow of the ChatGPT concept with selected features from the Google Gemini concept, particularly visible assistance controls and organized mistake feedback.

Step 1 — Start a Study Session

When students open Study Coach AI, they see three main options:

- Study a Topic
- Practice Exam Questions
- Review My Mistakes

Students select the task that matches their immediate learning goal.

Step 2 — Select Course Material

Students select a course and a topic or use a small set of course materials provided in the prototype. If supported by the implementation, they can upload a PDF containing lecture notes.

The chatbot should prioritize the available course material. If no relevant material is available, it must explain this limitation rather than implying that its response is based on the student's lecture.

Step 3 — Choose the Assistance Level

Before receiving help, students choose one of three assistance levels:

- **Hint:** Provides a clue or guiding question without revealing the complete answer.
- **Step-by-Step Explanation:** Explains the solution in manageable steps.
- **Full Answer:** Provides the complete answer with an explanation.

These options remain accessible during the study task. Students can change their choice whenever they need more or less help.

Step 4 — Interact with the Chatbot

The chatbot provides assistance according to the student's selected level.

During practice, students can attempt to answer independently, request a hint, ask for a step-by-step explanation, or reveal the complete solution. The chatbot should not automatically reveal the full answer when the student has selected a hint.

Step 5 — Review and Understand Mistakes

After submitting an answer, the student receives feedback based on the available reference answer, course material, or evaluation criteria.

When a likely mistake is identified, the chatbot should:

1. Identify the part of the answer that may be incorrect.
2. Explain the reason for the mistake when sufficient evidence is available.
3. Offer the student a choice between another hint, a step-by-step explanation, and a full correction.
4. Allow the student to revise the answer or attempt a similar question.

If the chatbot cannot confidently evaluate the answer, it should communicate this uncertainty instead of presenting an unsupported correction as a fact.

Step 6 — Adjust Preferences or Challenge the AI

Students can request shorter or more detailed explanations and indicate when an answer appears incorrect.

When a student challenges a response, the chatbot should reconsider it using the available course material or reference answer. It should acknowledge uncertainty when the evidence is insufficient and allow the student to provide additional context.

Step 7 — Complete the Study Task

At the end of the task, students choose whether to:

- Retry the question.
- Practice a similar question.
- Review the explanation.
- Return to the main menu.
- End the session.

This final step ensures that the AI does not force students into a fixed learning path.

5.5 Alignment with the HAI Principle: User Control and Agency

The selected HAI principle is **User Control and Agency**. This principle means that users should be able to make meaningful decisions about how an AI system behaves and how they interact with it.

Study Coach AI supports this principle through the following design decisions:

1. Choice and Decision-Making

Students choose their study goal, course material, assistance level, and next action. The chatbot supports their decisions instead of imposing a single learning process.

2. Adjustable Assistance

Students can move between Hint, Step-by-Step Explanation, and Full Answer without restarting the task. This allows them to determine how much support they need at different stages.

3. Transparency and Uncertainty Communication

The chatbot explains when relevant course information is missing or when it cannot confidently verify an answer. Students can then decide whether to provide more information, receive a general explanation, or skip the question.

4. Correctability and User Feedback

Students can challenge a potentially incorrect response, provide additional context, and request that the chatbot reconsider its answer. Their feedback becomes part of the ongoing interaction.

5. Reversibility and Freedom to Exit

Students can retry a question, change their assistance level, select a different task, or end the session at any time. They are not required to follow a suggestion or continue an interaction they no longer find useful.

6. Support for Independent Learning

The chatbot offers guidance at the level selected by the student. By making hints available without automatically revealing the full solution, it supports independent problem-solving while keeping more detailed help accessible when requested.

Together, these decisions make User Control and Agency a central part of the interaction design rather than a feature limited to a settings menu.

5.6 Planned Prototype Scope

The project will focus on developing a manageable web-based prototype that demonstrates the selected HAI principle through a small number of clearly defined interactions.

Core Features

1. Chatbot Interface

A simple interface where students can start a study task, view AI responses, submit answers, and navigate between interactions.

2. Study Task Selection

Three entry options: Study a Topic, Practice Exam Questions, and Review My Mistakes. The prototype may use a limited set of predefined tasks to keep development manageable.

3. Assistance-Level Controls

Three visible options: Hint, Step-by-Step Explanation, and Full Answer. The chatbot's response should reflect the selected option.

4. Practice Questions and Feedback

A small collection of verified questions from one selected course or topic. Students can submit answers and receive feedback appropriate to the question type.

5. Basic Mistake Explanation

The chatbot identifies likely errors and provides an explanation when the available reference information supports it. Advanced automatic classification of every possible error is outside the initial scope.

6. Preference Adjustment

Students can change the assistance level and request shorter or more detailed explanations during a session.

7. Uncertainty and Correction Handling

The interface includes a way to indicate that an answer may be incorrect. When course information is missing or insufficient, the chatbot explains the limitation and offers reasonable next steps.

8. Student-Controlled Next Actions

Students can retry a question, request another explanation, practice a similar question when available, or finish the session.

Implementation Boundaries

To ensure feasibility within the project timeline, the initial prototype will:

- Focus on one course or a small, clearly defined set of topics.
- Use an existing language-model API if available and permitted.
- Use predefined questions and reference answers where appropriate.
- Prioritize visible user controls and predictable interaction behavior over advanced AI capabilities.
- Avoid training a custom language model.
- Treat PDF upload and automated document retrieval as implementation-dependent features that will be included only if the team has sufficient time and resources.

The prototype is intended to demonstrate the interaction design and the selected HAI principle. It is not intended to replace a university learning management system or guarantee the correctness of every AI-generated response.

Initial Prototype Priorities

The initial prototype will prioritize the core interactions that demonstrate User Control and Agency:

1. A student can select a study task.
2. A student can choose between Hint, Step-by-Step Explanation, and Full Answer.
3. The chatbot responds according to the selected assistance level.
4. A student can submit an answer and receive basic feedback.
5. A student can challenge a potentially incorrect AI response.
6. The chatbot communicates when the available information is insufficient.
7. A student can retry the task or finish the session.

Advanced features, including detailed automatic error classification, persistent personalization, and a separate Socratic Mode, will remain optional until the team confirms that they are feasible within the available time and resources.

5.7 Proposed Evaluation

The prototype will be evaluated using a small set of predefined study tasks and test scenarios.

The evaluation will examine whether:

1. Students can identify and select the three assistance levels.
2. The chatbot follows the selected assistance level instead of automatically revealing a full answer.
3. Students can change their preferences without restarting the task.
4. Students can submit an answer and receive understandable feedback when a reliable reference is available.
5. The chatbot communicates uncertainty when course information is insufficient.
6. Students can challenge a response and choose what to do next.
7. Students can complete or exit a study task without being forced into a predefined path.

The team will use these scenarios to identify interaction problems and determine whether the prototype provides meaningful user control.

5.8 Expected Outcome

The expected outcome is a web-based Study Coach AI prototype that demonstrates how an educational chatbot can provide flexible study assistance while preserving student choice.

The prototype should show that students can control the amount of help they receive, respond to feedback, question potentially incorrect AI responses, and decide how to proceed throughout a study session.

The project will be considered successful if its core interactions consistently respect student choices and clearly communicate the limits of the available information.

5.9 Approval Request

We propose Study Coach AI as a student-controlled study assistant based on the HAI principle of User Control and Agency.

We request approval of the proposed interaction flow, selected features, and initial prototype scope before proceeding to detailed interface design and implementation.

We welcome feedback on the proposed design and any required changes to ensure that the project meets the course requirements and remains feasible for our team.